

Home Network Connectivity Report

The aim of this experiment was to understand how devices communicate over a local Wi-Fi network and to use networking tools such as Ping, Traceroute (Tracert), and ARP to test connectivity and identify connected devices.

Step 1

Connected the laptop and Android phone to the same Wi-Fi network.

Step 2

Checked the phone's IP address from the Wi-Fi settings.

192.168.1.105

Step 3

Opened Command Prompt and executed

ping 192.168.1.105

The laptop received replies successfully.

Step 4

Executed

tracert www.youtube.com

The route showed approximately **6 hops** between the laptop and YouTube.

Step 5

Disconnected the phone from Wi-Fi.

Run

ping 192.168.1.105

The command displayed

Request timed out.

showing that communication had failed.

Step 6

Executed

`arp -a`

to display the list of devices connected to the local network.

Results

- Ping test was successful while both devices were connected.
- Ping failed after disconnecting the phone.
- Traceroute displayed the path to YouTube through multiple routers.
- ARP displayed devices connected to the local network.

This report demonstrated how devices communicate within a Local Area Network (LAN). The **Ping** command was used to verify connectivity between two devices. The **Traceroute** command showed the route taken by packets to reach an Internet server, including the number of intermediate hops. The **ARP** command listed devices currently connected to the home Wi-Fi network. Overall, the experiment improved understanding of basic networking concepts such as IP addressing, packet transmission, routing, and device communication.